

CDTRS Development Setup Cheat Sheet

Development environment and installation guide.

1. Install

Python 3.12.x (python.org)
Git (git-scm.com)
VS Code (code.visualstudio.com)
PostgreSQL 16 + pgAdmin (postgresql.org)

2. Verify

```
python --version  
pip --version  
git --version  
psql --version
```

3. Project Setup

```
mkdir CDTRS  
cd CDTRS  
python -m venv venv  
venv\Scripts\activate  
python -m pip install --upgrade pip
```

4. Install Libraries

```
pip install PySide6==6.11.1  
pip install SQLAlchemy==2.0.38 psycopg2-binary==2.9.10  
pip install paddlepaddle  
pip install paddleocr  
pip install transformers torch  
pip install opencv-python Pillow  
pip install pymupdf  
pip install spacy  
python -m spacy download en_core_web_sm  
pip install rapidfuzz  
pip install pywin32  
pip install APScheduler  
pip install bcrypt  
pip install openpyxl reportlab  
pip install python-dotenv  
pip install numpy pandas  
pip install pyinstaller
```

5. Library Usage

PySide6: GUI; SQLAlchemy+psycopg2: PostgreSQL; PyMuPDF: PDF reading; OpenCV: preprocessing; PaddleOCR: printed OCR; TrOCR: handwritten OCR; Regex: structured fields; spaCy: NER; RapidFuzz: department/employee matching; pywin32: Outlook; APScheduler: reminders; bcrypt: password hashing; openpyxl/reportlab: reports; dotenv: config; numpy/pandas: utilities.

6. Pipeline

PDF/Image → PyMuPDF → OpenCV → PaddleOCR/TrOCR → Regex → spaCy → RapidFuzz
→ SQLAlchemy → PostgreSQL → Dashboard → Outlook

7. requirements.txt

```
pip freeze > requirements.txt  
pip install -r requirements.txt
```

8. EXE

```
pyinstaller --onefile --windowed main.py
```